ACM Forensic Audit of MOND-Favored Galaxies: Residual Pathologies in the SPARC Sample

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We present an ACM-based forensic audit of the galaxies that appear to favor MOND in the SPARC sample. The purpose of the paper is not to introduce yet another replacement function or empirical low-acceleration curve. It is to ask whether the MOND-favored regime represents a unified, stable, and reproducible physical mechanism, or whether it is better understood as a mixed residual regime shaped jointly by observational and modeling conventions. Paper II therefore adopts the retained ACM galaxy trunk established in Paper I and shifts the focus from global model comparison to full-sample residual pathology in SPARC [1].

Under a single ACM baseline applied to 164 comparison objects, the ACM-better systems form the dominant physical cloud, whereas the MOND-favored minority does not behave like a coherent dynamical counterexample. Instead, it separates into three distinct mechanisms: a geometry-sensitive subset that flips under distance-edge stress tests; a stellar-normalization-sensitive subset recoverable under conservative stellar mass-to-light rescaling; and a hard31 subset whose outer gas distributions are unusually flat, extended, and structure-poor. This decomposition suggests that the MOND-favored regime is better interpreted as a set of residual mechanisms stabilized under historical limitations, rather than as a unified physical stronghold.

The claim of this paper is not that earlier researchers committed deliberate errors. MOND's historical success was real and important as an empirical low-acceleration framework [2, 3]. Our claim is instead that, under earlier distance systems, stellar-normalization practices, and low-structure outer-disk regimes, a local threshold law could absorb multiple residual structures and thereby stabilize an interpretively overfit-friendly observational regime. We do not declare a final verdict on behalf of future data. We argue, however, that harder distance anchors, more stable stellar mass normalization, and higher-resolution outer-disk measurements will continue to arbitrate the interpretation of these regimes.

I. INTRODUCTION

A. Starting from the unified trunk of Paper I

The ACM galaxy-dynamics trunk is not introduced here for the first time. Paper I defined the retained galaxy trunk, including the unified constant family, the local background-response chain, the propagation-layer logic, and the global matched-sample performance of the stable trunk [4][1]. More importantly, Paper I fixed the methodological discipline that Paper II continues to follow: failed objects are retained rather than erased by per-galaxy patches, and explanatory power is not purchased by post hoc degrees of freedom.

Paper II therefore does not take on the task of re-deriving the first principles of the ACM galaxy trunk. All ACM calculations, residual classifications, and regime audits in this paper are performed on top of the single baseline already established in Paper I. The question here is not whether ACM works globally in the first place. The follow-up question is more specific: once a unified ACM baseline has been fixed, what does it mean that a minority of systems still appears to favor MOND? Do these systems form a coherent physical counterexample, or do they instead compress several distinct bias channels and fragile regimes into what only looks like a stable MOND stronghold?

B. From model comparison to residual pathology

Comparisons between ACM and MOND can easily remain at a coarse level: which model has the lower median residual, which model wins on more galaxies, and which model follows the low-acceleration empirical curve more closely. Such comparisons are useful, but they do not answer the specific question pursued in this paper. Once a minority of systems appears MOND-favored, repeating that ACM is globally stronger does not by itself explain what these galaxies actually represent.

This paper therefore shifts the object of study from simple model competition to residual pathology. We do not treat the MOND-favored subset as a naturally unified counterexample class. We treat it instead as a set of residual regimes that must be audited: are these systems geometry-locked, stellar normalization-biased, or so flat in their outer gas structure that any trunk relying on local structural activation is driven into an information-poor regime? In that sense, the central product of this paper is a full-sample pathology map rather than a ranked win–loss table. This shift is motivated not only by ACM's retained baseline, but also by the observational framing established by SPARC [1], MOND phenomenology [3], and the radial-acceleration literature [5].

This shift carries a deeper consequence. The aim here is not to put MOND on trial as a historical actor. MOND's historical success was real. It provided a courageous local approximation and, for a long period, served as a bridge between low-acceleration observations and theoretical tension. What this paper audits is how that local success became stabilized under historical conditions, and how its structural limits emerge once the sample is examined at higher residual resolution. It

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is in this restricted methodological sense that we use the word *overfitting*: not to suggest subjective manipulation, but to describe a regime-dependent structure stabilized under historical observational conditions.

II. AUDIT PHILOSOPHY AND UNIFIED ACM BASELINE

A. Why a unified baseline matters

All analyses in this paper are built on the stable galaxy trunk established in Paper I [4]. In practice this means that we do not refit ACM freely on a galaxy-by-galaxy basis, nor do we introduce new free parameters in order to rescue a small stubborn subset. We deliberately place ACM in a strict, unified, and reproducible audit posture: one trunk function, one constant family, and one local-response chain applied across the full comparison sample.

The purpose of this choice is not simply parameter economy. It is to prevent Paper II from collapsing back into empirical curve chasing. The goal is not to show that ACM can also behave like an old-style fitter with enough targeted adjustments. Rather, the goal is to test how stable the purported MOND-favored regime really is. If a subset remains MOND-favored only so long as the distance scale stays soft, the stellar mass normalization stays flexible, or the source family remains within a particular observational regime, then that apparent loyalty to MOND must itself become an audit object.

B. The four audit actions

To pursue this question, Paper II uses four mutually reinforcing audit actions. The first is the *distance-edge test*, in which the adopted distance is shifted within its quoted uncertainty, $D \pm e_D$, and the verdict response is measured. The second is the *stellar normalization scan*, in which the stellar mass-to-light ratio is varied within a conservative and astrophysically interpretable range. The third is the *outer-gas-shape diagnostic*, which uses outer-disk gas slope, curvature, and persistence to identify systems whose local structural signal is intrinsically weak. The fourth is the *source- and-family audit*, which tests whether distance-source families, reference chains, and observational regimes couple systematically to the residual mechanisms. These audit actions are motivated by the status of SPARC mass models [1], the LSB rotation-curve literature [6], stellar mass-normalization practice [7], and the distinction between soft and hard distance anchors such as the TRGB method [8].

All four actions share the same principle: none of them attempts to smuggle a new free parameter into the ACM trunk. The audit target is the sample and the interpretive regime, not the invention of a new decorative patch.

The comparison sample itself is not newly defined in Paper II. The 164-object set used throughout this audit is the same matched comparison set inherited from Paper I, with the same inclusion criteria and the same object-level baseline used for direct ACM–MOND comparison. Paper II therefore changes the mode of analysis, not the underlying sample definition.

C. From 62 to 22 / 9 / 31

The decomposition of the MOND-favored minority was not obtained by subjective inspection. It emerged from a clear audit sequence. First, the unified trunk was applied to the 164-object comparison sample, yielding an initial split into 102 ACM-better systems and 62 MOND-favored systems. Second, the 62-object subset was subjected to the distance-edge test. Twenty-two of these systems flip back to ACM under edge shifts, and the flips occur overwhelmingly on the $D+e_D$ side. The first audit layer therefore removes a geometry-sensitive subset from the supposed MOND stronghold. The full object-level audit table for this inherited comparison set is released with the evidence package associated with Paper II.

Third, the remaining holdout systems were subjected to a conservative stellar mass-normalization scan. Within the residual 40-object pool, nine systems return to ACM under plausible M/L adjustments. Only after these first two filters does the final hard31 subset emerge. The three mechanisms discussed in this paper are therefore not preconceived labels; they are the outcome of a stepwise sequence of distance-prior audit, stellar-normalization audit, and outer-structure/source-regime audit.

D. Minimal mathematical definitions

Although this paper does not re-derive the ACM trunk, the audit procedure must remain mathematically reproducible. All comparisons in this paper inherit the unified retained trunk of Paper I, whose dynamical closure is written as

$$g_{\rm ACM}(r) = g_{\rm bar}(r) + \frac{2}{3}c^2\eta_{\rm local}(r), \tag{1}$$

where $g_{\rm bar}(r)$ is the baryonic acceleration and $\eta_{\rm local}(r)$ is the retained local background-response field from Paper I [4]. Paper II does not repeat the first-principles derivation of $\eta_{\rm local}$; it inherits it as part of the single audit baseline.

On top of that baseline, only a few additional mathematical definitions are introduced. The distance-edge test is implemented as

$$D_{\rm new} = D \pm e_D, \tag{2}$$

with radius and baryonic components rescaled consistently under the shifted distance. The stellar normalization audit is written as

$$\Upsilon_\star \to s\,\Upsilon_\star, \tag{3}$$

where s is restricted to a conservative and astrophysically interpretable range.

For the hard31 structural audit, the relevant quantities are not new dynamical equations but diagnostic structure measures. The outer-disk gas slope is defined by

$$S_{\rm gas}(r) = \left| \frac{d\ln(V_{\rm gas}^2)}{d\ln r} \right|, \tag{4}$$

and the outer-disk gas curvature by

$$C_{\rm gas}(r) = \left| \frac{d^2\ln(V_{\rm gas}^2)}{d(\ln r)^2} \right|. \tag{5}$$

These quantities are not treated as new corrective theory terms. They are used as structural fingerprints that distinguish systems with genuinely weak local structure from systems that remain information-rich but dynamically difficult.

III. GLOBAL RESIDUAL PATHOLOGY MAP

A. The 164-object pathology atlas

To move from anecdotal cases to sample-level structure, the full 164-object comparison sample is projected into a residual pathology map. The result is clear. The ACM-better 102 do not form a marginal survival group; they form the dominant physical cloud of the sample. By contrast, the MOND-favored minority does not behave like a single physical family. It separates into three distinct mechanisms.

The geometry-hostage 22 occupy the high-distance-error corner and are enriched in soft distance sources. The stellar-hostage 9 inhabit the low-stellar-light, gas-rich sector, but can be restored by conservative stellar mass normalization. The gas-flat hard31 form a low-structure regime characterized by flat outer gas profiles, weak curvature, high outer persistence, and the absence of hard geometric distance anchors. In simple numerical terms, the sample becomes 164 objects decomposed into 102 ACM-better systems and 62 apparent MOND-favored systems, which in turn separate into 22 geometry hostages, 9 stellar hostages, and 31 hard31 judges.

B. A four-quadrant verdict

The four-quadrant pathology map yields a clear verdict. In the geometry plane, the geometry-hostage 22 cluster in the high-error sector and are the only class that shows a systematic verdict reversal under $D + e_D$. In the mass/baryon plane, both the stellar-hostage 9 and the gas-flat hard31 occupy low-stellar-light, gas-rich territory, but only the former return under conservative M/L recovery. In the outer-gas-shape plane, the hard31 concentrate in the low-slope, low-curvature, high-persistence corner. They are not gas-free systems. They are systems whose gas support is too flat, too extended, and too poor in local structure to provide strong local feedback.

The importance of the pathology map is that it dissolves the MOND-favored region into multiple residual mechanisms. The subsequent sections are therefore not arbitrary case studies; they are successive unfoldings of the full-sample pathology map.

IV. GEOMETRY HOSTAGES

The distance-edge test produces one of the sharpest signals in the paper. Among the 62 MOND-favored systems, 22 flip to ACM once the distance is moved to the edge of the quoted uncertainty range. These reversals occur overwhelmingly on the $D + e_D$ side rather than symmetrically on both sides. The median relative distance error of this subset is approximately 24.96%, and its internal fraction of $f_D = 1$ systems is about 77.3%, both substantially above the level of the dominant sample cloud.

These signals support a simple interpretation: the geometry-hostage 22 are not natural MOND victories. They are systems whose verdict is being locked by the current geometric scale. Their apparent MOND advantage does not require a new dynamical regime. It requires only a slightly compressed distance prior capable of holding a local threshold law in an artificially favorable position. In that sense, they are better interpreted as geometry problems than as dynamics problems.

V. STELLAR HOSTAGES

The second audit layer reveals a smaller subset with strong sensitivity to stellar mass normalization. When the remaining holdout systems are scanned under conservative M/L rescaling, 9 of the 40 return to ACM. The most common recovery scale appears on the higher but still plausible end of the allowed range. These systems therefore do not require a new dynamical law. They behave instead like systems that have been held on the MOND-favored side by a weak stellar normalization anchor, consistent with the long-standing ambiguity in disk-galaxy mass-to-light estimates [7].

This class shares a low-stellar-light, gas-rich background with the hard31, but it does not share the same pathology. The stellar-hostage 9 are not entirely structure-poor. Their key feature is that they are especially sensitive to the normalization of the stellar component. They are therefore closer to recoverable normalization bias than to a genuinely separate low-acceleration dynamical regime.

VI. THE HARD31 AS INTERNAL AUDIT JUDGES

The status of the gas-flat hard31 is categorically different from that of the first two groups. These 31 systems were not initially selected as ammunition against MOND. They were first used as an internal ACM audit benchmark. In other words, the first suspicion was not that the data were biased, but that the ACM trunk might still be missing a sufficiently natural, as-yet-unmerged correction.

For precisely that reason, the hard31 became the most severe test set. A series of physically natural and zero-new-

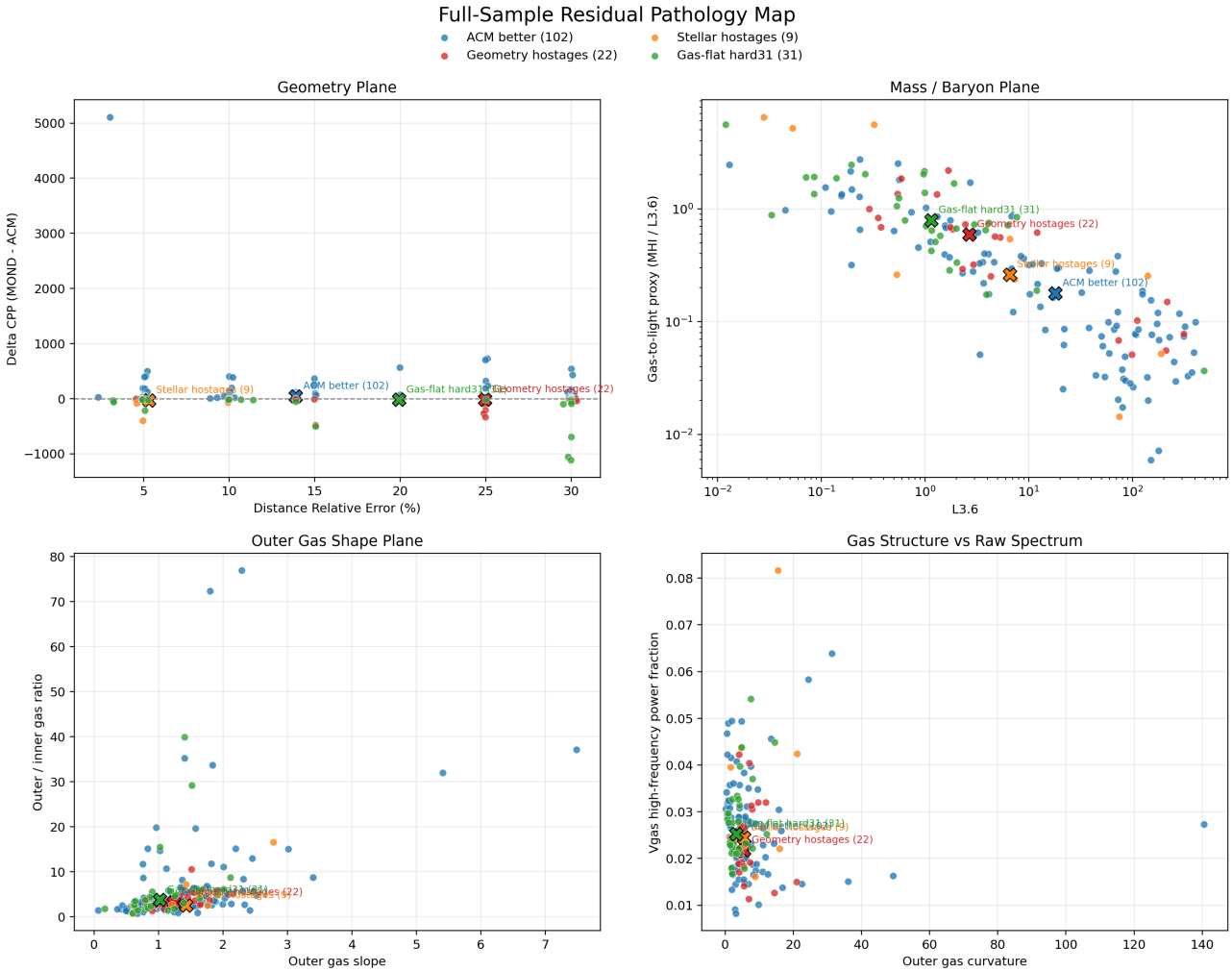


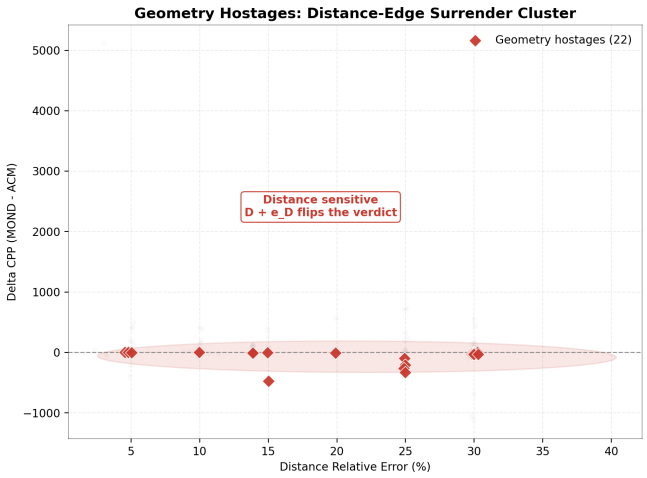
FIG. 1. Full-sample residual pathology map for the 164-object comparison set. The ACM-better cloud serves as the primary physical trunk, while the three MOND-favored residual regimes—geometry hostages, stellar hostages, and gas-flat hard31—are isolated for forensic auditing.

parameter operator variations was explored: first-gradient schemes, second-curvature schemes, diffusion accumulation terms, self-shielding terms, and nonlinear impedance couplings. The associated raw experimental notes and archived negative trials will be released progressively as research records at an appropriate stage. The issue is not that these ideas were directionless. The issue is that none of them materially changed the fate of the hard31. They were either too weakly activated or only produced small perturbations. None was sufficient to bring the systems back across the verdict boundary.

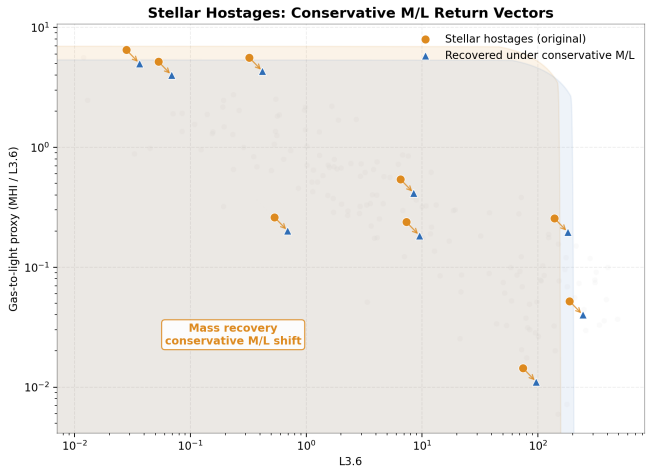
More importantly, the hard31 exhibit a coherent set of quantitative fingerprints. Their number of hard geometric anchors is zero ($f_D = 5 = 0$). Their mean outer gas slope is approximately 1.12, their mean outer gas curvature is approximately 3.96, and their mean outer-to-inner gas persistence ratio is approximately 5.95. In mechanical terms, such systems are close to an "idle-running" state: the outer disk is so flat, extended, and poor in structure that insufficient structural variation remains to generate stable feedback. Any model that depends on local-response activation will therefore enter a weak-coupling regime here. For that reason, the hard31 do not resemble a simple pile of bad points. They instead define a class of soft-source, low-anchor, low-structure systems occupying a special observational regime.

This is the turning point of the paper. Once all naturally favorable ACM-side operator modifications fail systematically, the problem is forced to move from "what function is still missing from the model" to "whether the input system itself carries historical bias." This is why the hard31 deserve the name *judges*. They are neither MOND's hostages nor ACM's enemies. They are the methodological subset that forces the audit to move from trunk-side ornament to input-side diagnosis.

The positive role of the hard31 is reinforced by their source topology. Their reference-family audit shows that they do not inhabit a hard geometric control group. They remain concentrated in softer source regimes and lack hard distance anchors entirely. This does not by itself prove input-side bias, but it does establish a strong regime-level audit concern. In that sense, hard31 is better treated not as a clean MOND fortress



Left: Geometry-hostage audit. Focused audit of the 22 systems that return to ACM under distance-edge stress tests. The systematic flip on the $D + e_D$ side identifies these objects as geometric artifacts rather than dynamical outliers.



Right: Stellar-hostage audit. Recovery trajectory for the 9 stellar-normalization-sensitive systems. Orange points mark the initial MOND-favored states; blue triangles denote their return to the ACM baseline after conservative M/L auditing.

FIG. 2. Sequential audit of the recoverable MOND-favored subsets. The left panel isolates the geometry-sensitive systems removed by the distance-edge test, whereas the right panel isolates the stellar-normalization-sensitive systems recovered under conservative M/L auditing.

but as a soft-source observational island that must remain in the main text rather than being relegated to supplementary material.

VII. DISCUSSION

Taken together, the results support a restrained but sharp reinterpretation: the MOND-favored regime should no longer be discussed as a unified physical stronghold. It behaves more like a distance–mass–model trap, or more broadly like an overfit-friendly observational regime stabilized under historical conditions.

This interpretation should not be read as an accusation of deliberate error. MOND's historical success was real. Under the observational conditions of its time, it provided a highly insightful local approximation and helped drive the field's re-engagement with low-acceleration phenomena. The point of the present audit is different. When the distance source is softer, the stellar mass normalization more flexible, and the outer disk flatter and poorer in local structure, a local threshold law can absorb several distinct residual structures and compress them into an apparently unified success region.

In that sense, *overfitting* is not used here as a moral category. Nor is it meant in the narrow sense of a merely over-parameterized fit. It is a methodological diagnosis at the interpretive and regime level. It describes a structure in which multiple residual regimes that should not belong to the same dynamical category become compressed, aligned, and stabilized under historical observational conditions. The audit of the 22, the 9, and the 31 is precisely the process of decomposing this stabilized overfit regime.

VIII. CONCLUSION

This paper is not written as a final declaration. We do not claim that ACM has already conquered all low-acceleration systems, nor do we reduce MOND's historical role to an erroneous episode. What the paper does establish is that the interpretive authority of the supposed MOND stronghold can no longer rest on a single label. It must instead be redistributed across distinct residual mechanisms: geometry-locking, stellar-normalization bias, and low-structure outer-gas regimes.

The main conclusion is therefore not "who has already won." It is that future discussions of the MOND-favored regime cannot be based on fit quality alone. They must first pass through audits of distance, mass normalization, outer structure, and source regime. The contribution of Paper II is to define that objective audit framework.

This also means that harder distance anchors, more stable stellar mass normalization, and higher-resolution outer-gas structure observations will do more than supplement the present paper. They will continue to arbitrate the interpretive ownership of these regimes. The hard31 in particular should not be treated as a safe harbor for MOND, but as priority targets for future higher-resolution observations and harder geometric distance determinations. We do not pronounce a verdict in advance on behalf of future data. The present audit nevertheless suggests that, as those data become harder, discussion of the MOND-favored regime will become progressively less sustainable at the level of surface fit quality alone and progressively more dependent on unified baselines, residual pathology, and regime-level audit.

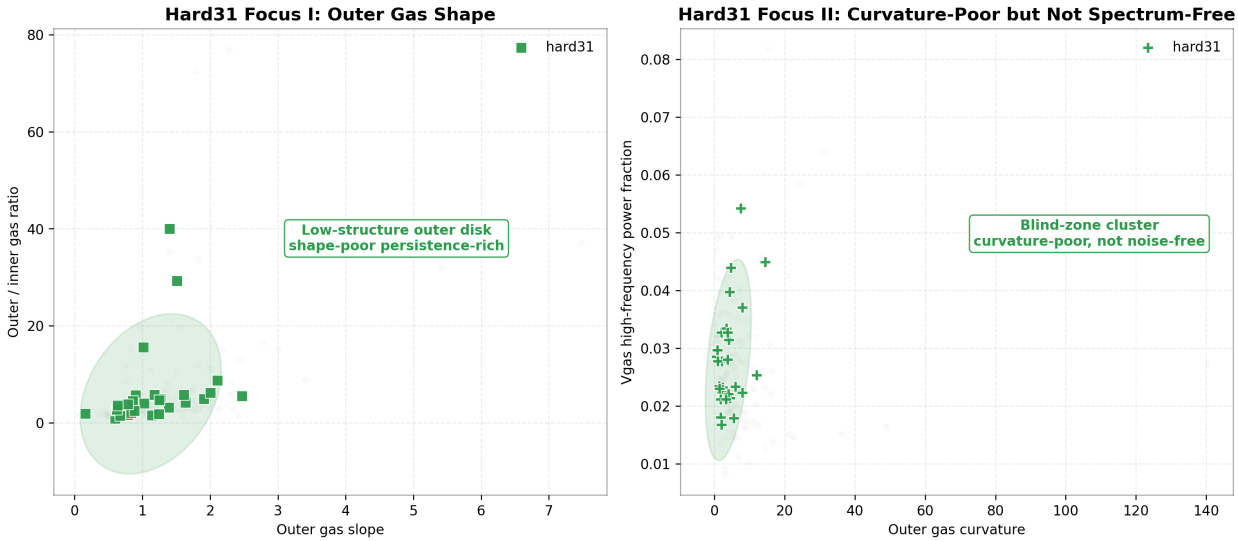


FIG. 3. Pathological fingerprinting of the hard31 subset. The subpanels isolate the low-slope, high-persistence outer-gas regime, demonstrating that these systems occupy a structure-poor observational blind spot.

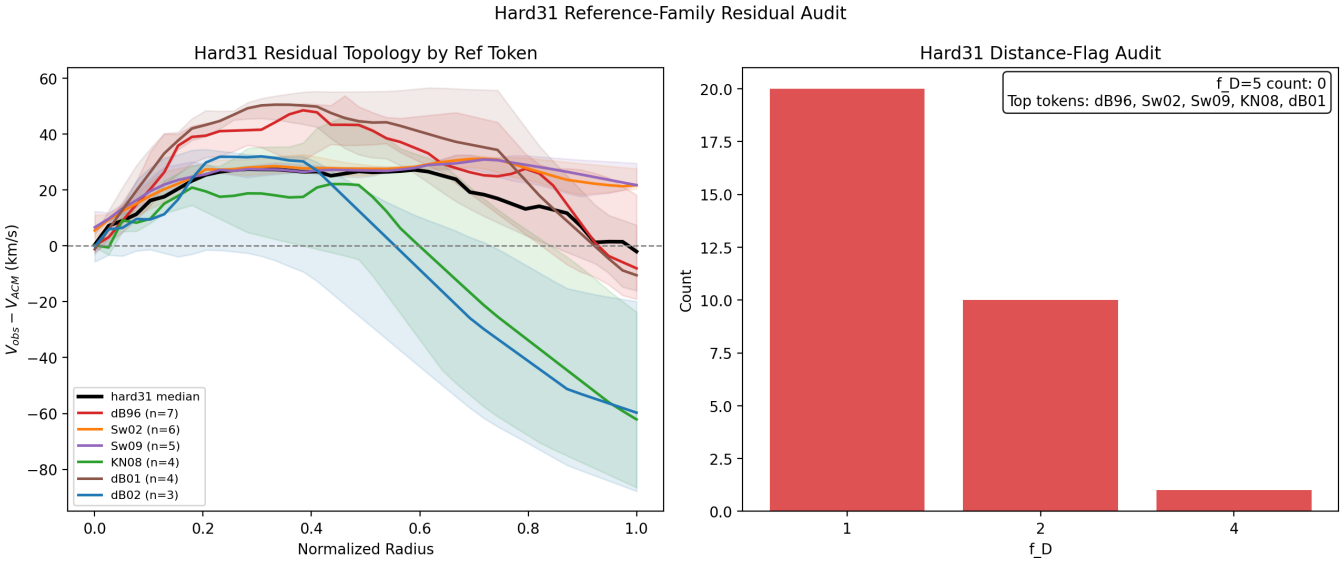


FIG. 4. Distribution of the hard31 subset within source-regime space. The concentration in softer reference families and the lack of hard geometric anchors support the interpretation of hard31 as a data-regime island rather than a unified physical counterexample.

IX. OBJECTIVE VALIDATION PATHS FOR FUTURE OBSERVATIONS

If the conclusions of this paper are to enter a genuinely reproducible phase, future work should not remain at the level of swapping empirical fit curves. It should instead proceed along three objectively higher-resolution paths. The first is harder distance anchoring. Systems that remain in soft distance-source regimes and exhibit strong step-like verdict responses under $D+e_D$ should be prioritized for TRGB, Cepheid, maser, or comparable hard geometric distance measurements [8]. The purpose is not to privilege any theory in advance, but to stabilize the geometric coordinate system itself.

The second is more stable stellar mass normalization. Systems with low $L_{3.6}$, high gas-to-light ratios, and strong sensitivity under conservative M/L scans should be re-examined under tighter stellar-population, color, and evolutionary constraints. The purpose is to reduce the phase space that can be absorbed by flexible normalization. The third is higher-resolution outer-disk structure measurement. For hard31-like systems—low slope, low curvature, high outer persistence, and no hard geometric anchor—the priority is not another local corrective function. The priority is improved H I outer-disk structure, more reliable inclination fields, and better constraints on outer-disk asymmetry.

The paper therefore closes not with a demand that future observers accept a particular theoretical conclusion, but with a validation program that any position should accept. First harden the distance scale. Then stabilize the stellar normalization. Then complete the outer-disk structural information. Any theory that wishes to continue treating the MOND-favored regime as a unified physical stronghold should first be required to pass through those same higher-resolution audit channels.

X. DATA AND CODE AVAILABILITY

The theoretical baseline used in this paper is inherited from Paper I, whose preprint is available via Zenodo DOI and whose public code repository is available at anchor-collapse-model. The public release associated with the present paper, including the bilingual audit frontend, the retained paper-II reproducible core, the curated evidence package, and the archived audit-process scripts, is prepared at galaxy-audit-system. At the time of writing, this second repository is being prepared for public release and is intended to serve as the public codebase for the audit layer developed in Paper II. It is also intended to host an interactive audit interface as a public-facing visual companion to the present manuscript. The final public web entry point will be added upon publication release.

Appendix A: Archived Audit Operators

To prevent the main text from being overwhelmed by exploratory corrections, we do not merge the full operator archive into the narrative body of the paper. This omission should not be read as methodological irrelevance. On the contrary, the hard31 subset acquires its status as a methodological judge because a sequence of physically motivated, no-new-parameter operators was tested against it and archived.

At minimum, this archive includes first-gradient corrections, second-curvature corrections, diffusion-accumulation terms, self-shielding terms, and nonlinear diffusion–shielding impedance couplings. All of them obey the same retained-trunk discipline: no per-galaxy freedom, no ad hoc rescue parameter, and no abandonment of the unified baseline. In each case the question is simple. Can a physically natural modification materially alter the hard31 verdict without compromising the stability of the ACM-better majority?

The archive verdict is clearer than any one operator formula considered in isolation. The tested operators were either too weakly activated to change the hard31 classification, or they produced only marginal perturbations while preserving global trunk safety. Their significance therefore lies not in having nearly succeeded, but in forming a reproducible body of negative results. This negative body of evidence moves the audit away from the repeated search for one more elegant local function and toward the diagnosis of historical bias on the input and regime sides.

In the final public form of the paper, the closed expressions, quantitative summaries, and archive verdicts of these operators can be documented in a longer appendix or linked research archive. For the present manuscript, it is sufficient to note that the hard31 subset remained stable against a broad family of no-new-parameter ACM-side rescue attempts. That stability is part of the reason the subset functions as a methodological judge rather than a generic collection of residual outliers.

Appendix B: Supplementary Regime Diagnostics

Not every diagnostic figure belongs in the main text. Some are more useful as exclusionary evidence or regime-audit evidence than as headline figures. In particular, the raw $V_{\rm gas}$ spectrum comparison between hard31 and the ACM-better majority is retained primarily to clarify what the hard31 subset is *not*. It is not merely a visibly fabricated frequency regime. Nor is it reducible to a trivial smoothing artifact.

Likewise, the distance-family summary tables, ranked distance-edge response lists, stellar-normalization sensitivity tables, and the hard31 geometry audit summary remain important to reproducibility even when they do not serve as the central visual anchors of the paper. Their function is not to amplify the rhetoric of the main text. Their function is to keep the paper's stronger claims traceable to explicit procedural audits. They can then be checked item by item by external readers.

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